

CAT® PAVING PRODUCTS

GUIDE TO SOIL COMPACTION

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FORWARD

Earth materials have always been the world's most abundant construction material. From the days when Romans used earth materials to build roads to modern-day dam construction, proper compaction operations have been linked to the performance of fills and to what is ultimately built on top of those fills. Today, projects such as roadways, embankments, dams, structural fills, and slopes require controlled compaction operations to ensure compliance with engineering quality criteria. When compaction quality is not achieved during construction, repair costs over performance life can exceed initial construction costs.

Earth materials are not just the most abundant construction material, but also the most variable. Particle size and shape, mineralogy, moisture content, mixings and time all contribute to the variability of earth materials. This guide describes sources of variability and focuses on selecting the correct equipment to maintain effective, productive compaction operations. Practices for selecting compaction equipment and earth material placement processes have been based largely on rules-of-thumb or what has worked in the past. However, with accelerated construction schedules and shifts toward results-oriented performance specifications, compaction operations—even though they are a small part of overall construction costs—are increasingly critical for project success.

This guide describes the basics of compaction processes, selection of effective compaction machines, and emerging quality assessment testing technologies with the goals of optimizing compaction time and controlling project costs. Guidance for machine selection is based on material type, target lift thickness, productivity needs and quality assessment criteria requirements (e.g., relative compaction, modulus of subgrade reaction).

This guide is particularly useful for contractors who want to gather and use soil information before projects start so they can develop site-specific bid prices and compaction process plans that focus on getting jobs done right the first time, avoiding costly rework and construction delays.

Compaction measurement technologies, like Cat® Compaction Control, now make it possible to monitor compaction progress in terms of lift thickness, pass coverage, Compaction Meter Value (CMV) and Machine Drive Power (MDP). CMV is a well understood parameter that allows







compaction machines to be measurement devices that provide mechanistic parameter values for soils (e.g., stiffness, strength) based on machine-ground vibration analysis. MDP is a new, innovative technology from Caterpillar that pushes the capabilities of these systems even further.

By measuring mechanistic values of compacted materials in real-time using GNSS (Global Navigation Satellite System) position information, color-coded quality assessment maps can be generated. These maps can be linked to design values to ensure that compacted materials meet established quality criteria. This approach to compaction quality assessment is light years ahead of the old method of visually inspecting for "walkout." Where compaction specifications previously were oriented toward method specifications that dictated the process, results-oriented performance specifications are now possible with 100 percent coverage in real-time using integrated technologies. In fact, intelligent compaction technologies are the basis for rethinking compaction quality assessment by agencies around the globe.

This guide combines experience and knowledge gained from Caterpillar's commitment to improving compaction equipment and operations. Many practicing contractors, engineers, recognized authorities and researchers contributed to this guide. Emerging compaction technologies, especially integrated intelligent compaction monitoring and methods for forecasting operation parameters, will drive the most significant changes in compaction assessment since 1933 when Proctor established standards for moisture control.

Users of this guide will have at their fingertips a practical resource for soil compaction principles, expert information about machine selection, and how to optimize compaction operations. Implementing the information in this guide will reduce risk and improve compaction quality.

David J. White, Ph.D.
Associate Professor
Iowa State University








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INTRODUCTION

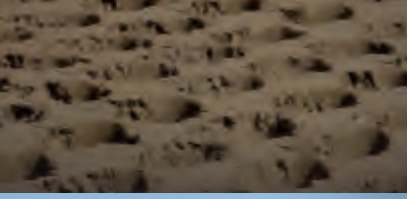
Those who want to build must compact.

Caterpillar is pleased to present this *Guide to Soil Compaction*. It is intended as a guide to soil compaction principles, testing techniques and on-the-job procedures. The content provides a practical approach to a rather complex theoretical subject. It is the product of decades of experience in the earthmoving industry, as well as the knowledge contributed by countless individuals who have worked with Caterpillar over the years.

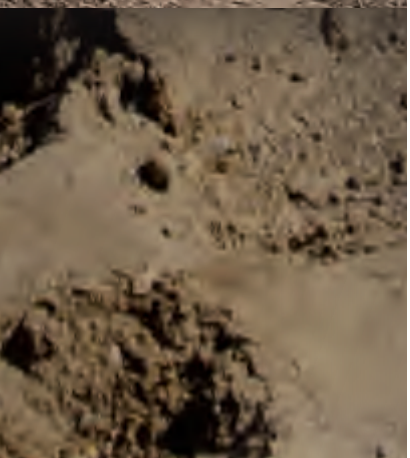
Whether you are a construction professional, a governmental official, an educator or student, or whether you are simply interested in learning more about construction methodology, you'll find this guide to be a valuable resource.

Another valuable resource to consult for earthmoving or compaction applications is your local Cat dealer. Dealer personnel are trained by experts from Caterpillar to assist you, providing the equipment, services and knowledge to keep you at your productive best.


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Unit 1
THE BASICS OF SOIL COMPACTION

The ability to analyze soil composition is critical to the process of establishing compaction specifications and then achieving the required load-bearing strength.

[WHAT IS COMPACTION?]

In simple terms, compaction is the process of mechanically increasing the density of a material. Soil is made denser by reducing the voids between the particles that make them up. In time, loose material settles and compacts itself naturally. By applying various mechanical forces, the time required to achieve compaction is shortened from years to hours.

Compaction is a necessary process for nearly every kind of construction project, including roads, railroads, airfields, building sites/ foundations, pipelines, dams, canals, culverts and more. If soil is required to support a structure, compaction usually is necessary to keep the structure stable.

Soil compaction is accomplished by employing one or a combination of forces: static pressure, impact, manipulation and vibration.





[WHY IS COMPACTION IMPORTANT?]

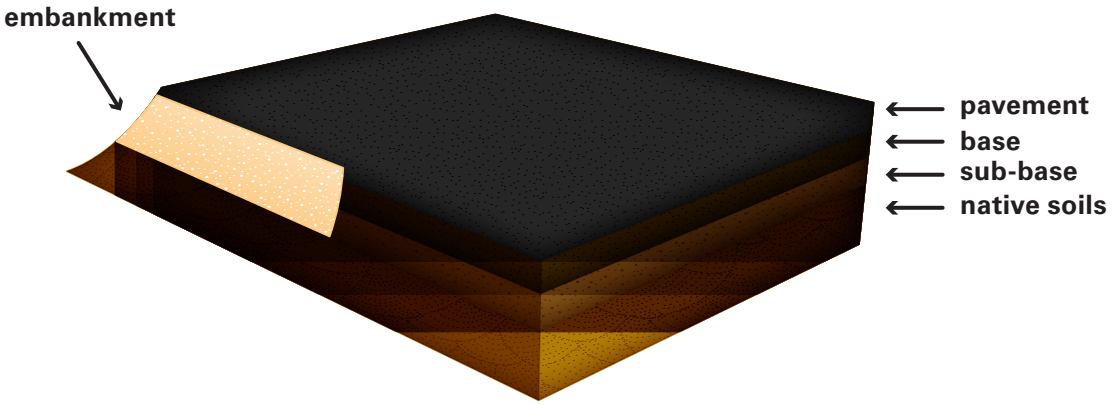
Properly compacted material is able to support heavier loads without deforming (bending, cracking, moving). The substrate material supporting a heavy structure must be very dense or it will compact even more under load, causing the structure to settle. Dense material has less permeability, which reduces the impact of water infiltration. Compaction also levels the surface and reveals structurally weak areas.

A good way to illustrate the importance of compaction is to consider the various layers of a typical roadway. Each layer of the roadway is designed to provide a specific engineering purpose, as well as support the weight placed on it. Each layer must be constructed of the right material to the proper thickness and stiffness. If one layer is not strong enough, the road will fail.

Compaction occurs during all phases of road construction. The quality of compaction has a significant impact on the longevity of the road, as well as a profound effect on the comfort—and possibly the safety—of the public who use it.

The least expensive element in extending the service life of a road is the compaction process. Increasing the density of the roadway layers during the construction process costs very little per cubic meter or yard of soil. Meeting density specifications can save significant money in future road maintenance and/or resurfacing costs.

TYPICAL SECTION OF ROADWAY



[WHAT IS SOIL?]

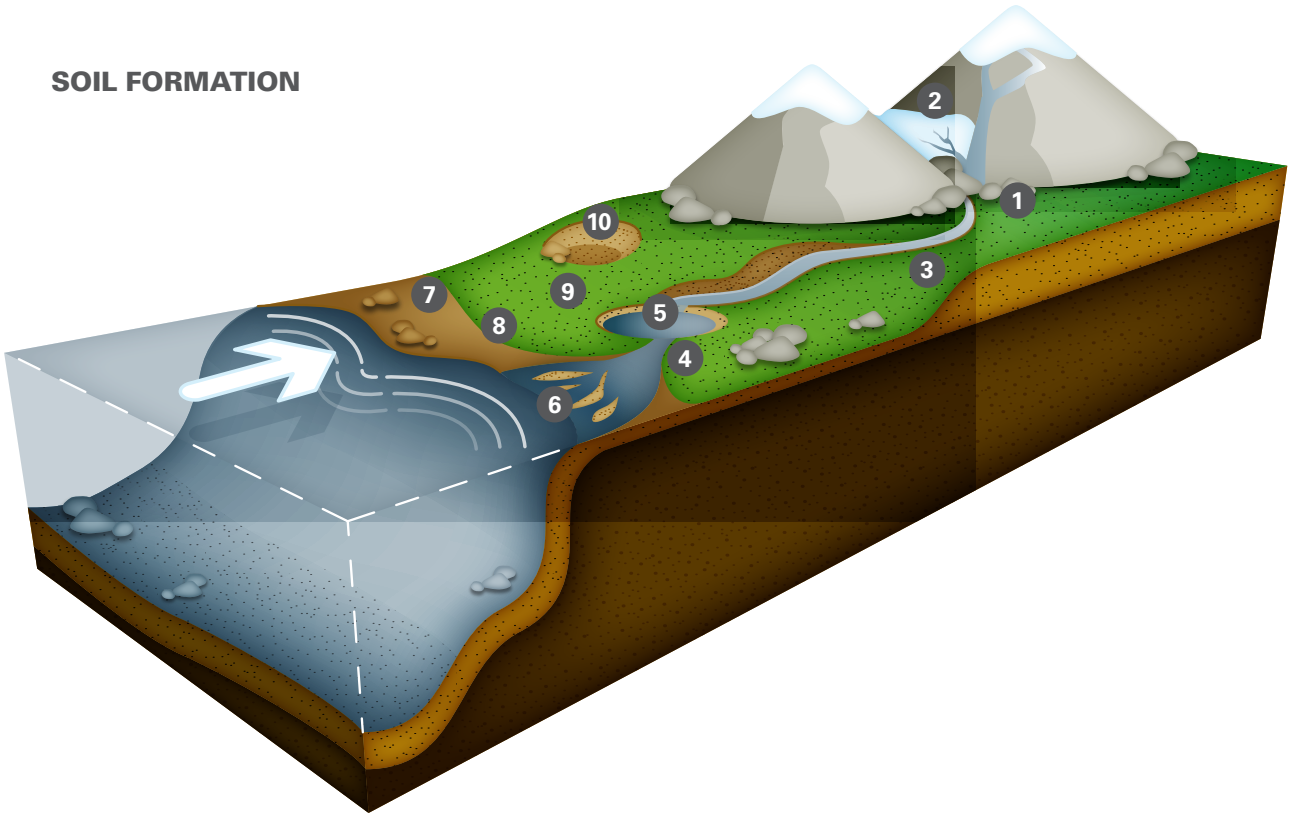
Soils are unconsolidated materials composed of mineral particles that may contain organic substances. Soils are essentially deposits of disintegrated rocks that have been slowly broken down by physical and chemical processes.

The physical processes include freezing and thawing, rolling, grinding and blowing.

Chemical processes form clay soils. Long-term weathering action and rainfall play an important part in creating clays. Clay differs from sand and gravel in that it consists of tiny, flat particles with plate-like structures that come from a variety of rocks.

Organic matter also contributes to soil formation. When plants die, their residue becomes part of the soil. Soils with high organic matter content are usually too spongy and weak to be used for structural purposes.

SOIL FORMATION



1. **residual soils**
   broken down pre-existing rock
2. **glacial deposits**
   materials carried or created by ice sheets
3. **glacifluvial deposits**
   materials transported by melt water from ice sheets
4. **river deposits**
5. **lake sediments**
6. **alluvial soils**
   fine-grained soils deposited on plains and in estuaries by flowing water
7. **wave-washed sediments**
8. **wind-blown deposits**
9. **organic soils**
   decomposed vegetation
10. **man-made soils**
    processed by blasting and crushing

[SOIL TYPES]

Although soils may vary widely in physical and chemical make-up, six fundamental types are recognized for engineering purposes: boulder, cobble, gravel, sand, silt and clay.

The six soil types are generally organized by particle size, which is determined utilizing a sieve test. While the precise details of engineering use can vary by country, the sieve sizes are generally defined by systems developed by one of two sources: the International Organization for Standardization (ISO) *[www.iso.org]* or the American Society for Testing and Materials (ASTM) *[www.astm.org]*. The two systems do not agree precisely, but they are very similar. Percentages of soils that are too fine to be sorted with sieves are determined by utilizing a hydrometer test. *(See page 14.)*

SOIL TYPES



Boulder



Cobble



Gravel



Sand

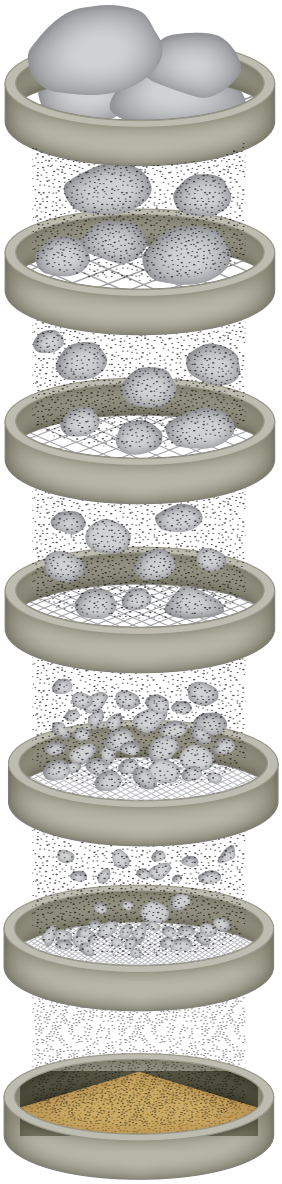


Silt

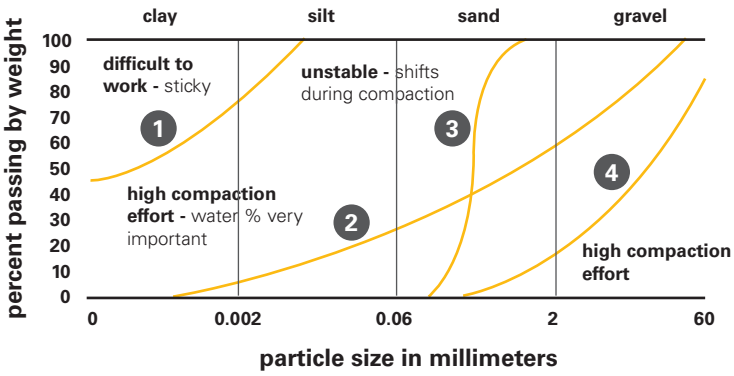


Clay

SIEVE TEST



PARTICLE SIZE DISTRIBUTION



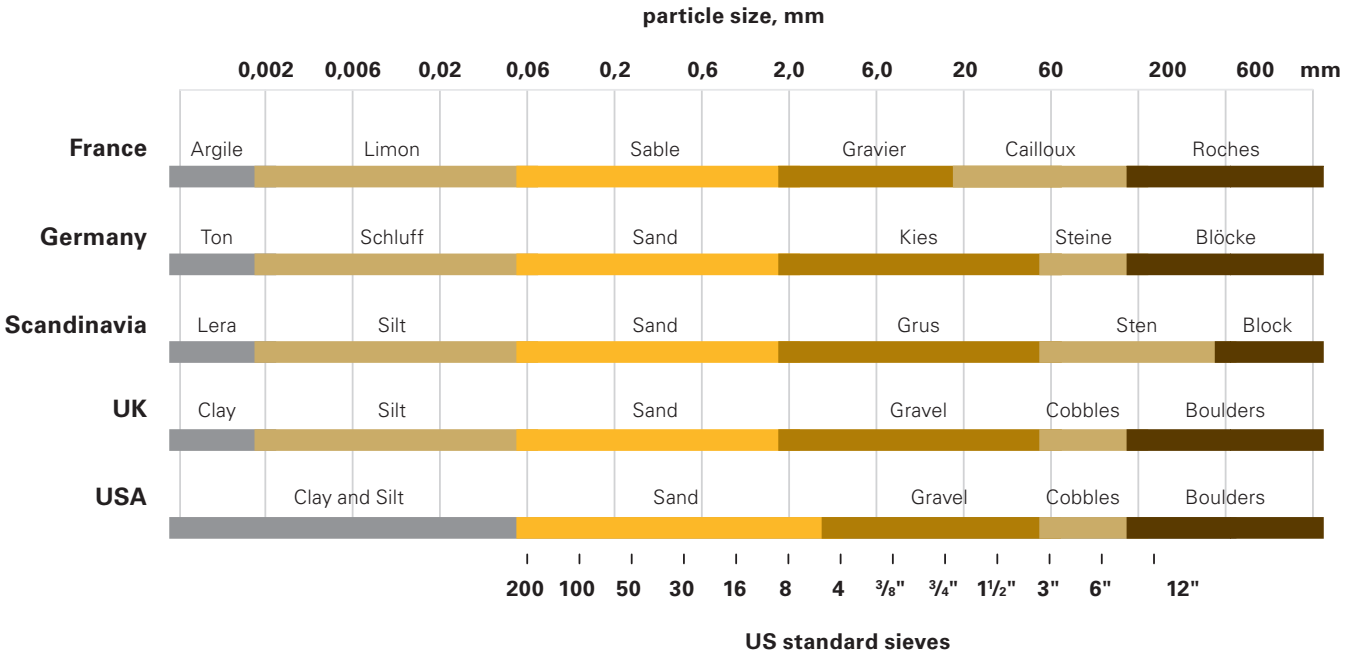
1. silt clay
2. gravel - silt
3. uniform sand
4. gravel - sand



Hydrometer Test

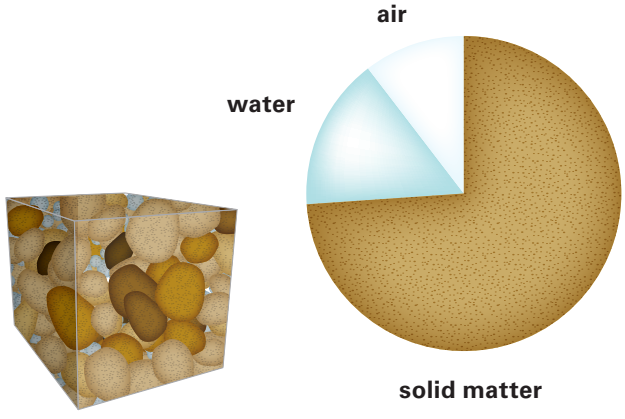
The soil sample is dispersed (suspended) in water in a graduated cylinder. The time it takes for the material to settle to the bottom can identify different grain sizes. A hydrometer reading of the suspension is taken to determine the specific gravity, which allows the percentages of grains for a particular size to be calculated.

SOIL COMPARISON CHART



A soil material is fundamentally a mixture composed of varying percentages of the previously mentioned soil types. It is important to understand that a soil material is not made of solid materials only. A soil material is a mixture of solid matter (any combination of soil types), water and air.

SOIL MIXTURE

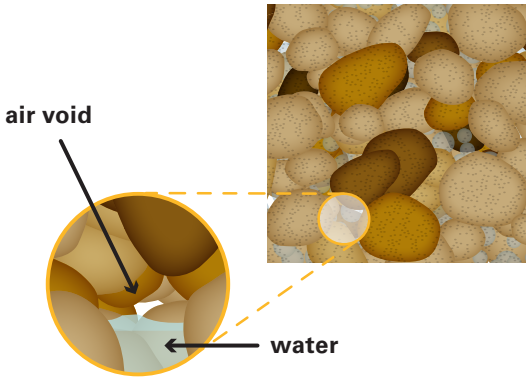


Natural or "native" soils—those that lie naturally deposited in the ground—will vary from one location to the next. For example, a soil material is never 100 percent clay or sand—there are always small percentages of other soil types present. The best soil materials for construction are often composites of specific percentages of choice soil types, depending on the engineering characteristics that are desired.

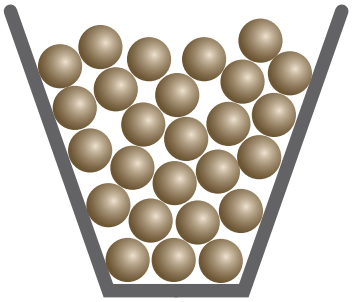
Geotechnical engineers can design soil mixes to provide these characteristics by specifying the percentages of each soil type. Soil types missing from a native soil are added in proportion and mixed with the natural soils to create an engineered soil. These additive materials are often determined by conducting an economic analysis of soil materials available nearby.

When you examine a soil sample closely, the individual particles are of many sizes and shapes. The spaces between the particles are called "voids." Voids can be occupied by either air or water. When a soil material has too much air and water due to abundant voids, the soil will be unstable. The process of compaction rearranges the soil particles to minimize the volume and size of air voids, making the material more dense and stable.

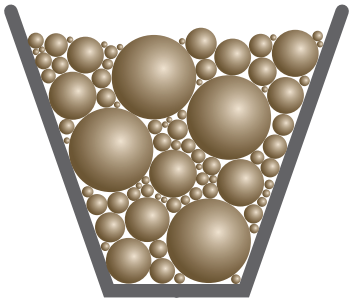
AIR/WATER VOIDS



GRADATION



uniformly or poorly graded



well graded

The individual particles of a soil material will vary in size, even if the variance is slight. The range of particle sizes is referred to as "gradation," or sometimes called "particle-size distribution" or "grain-size distribution." Ideally, there are relatively equal amounts of all grain sizes with no predominant size present. A material that possesses this ideal range of sizes is referred to as "well-graded."

Materials that possess particles of nearly identical sizes, such as alluvial sands, are "uniformly graded" or "poorly graded" materials. If one or more sizes are not present in the material, it is called "gap-graded." A well-graded soil will compact easier than a poorly graded soil because the presence of grain-size variability means the smaller grains fit nicely in the voids between the larger grains.

[FOUR TYPES OF SOIL MATERIAL]

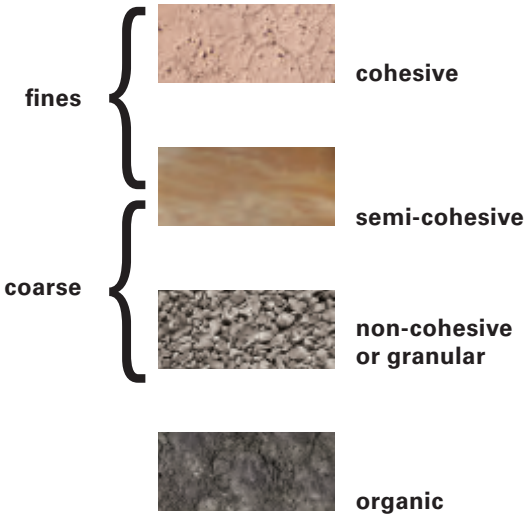
While it is sometimes useful to know the exact composition of a soil material, it is more pragmatic to understand how a soil material reacts when various forces are applied to it. To that end, professionals who work with soil classify soil materials into four basic types:

1. **Cohesive**
2. **Semi-cohesive**
3. **Non-cohesive**
4. **Organic**

Each type reacts differently when forces are applied to it. How each reacts dictates its suitability for various engineering purposes in construction, as well as what kinds of means are used to work with each soil. As previously mentioned, organic soil materials are not suitable for construction purposes.

When a soil material is unsuitable for engineering purposes, it is either replaced or various means are employed to improve the characteristics of the soil, a process referred to as "stabilization." These solutions can include chemical stabilization—for example, the incorporation of Portland cement, lime, fly ash or calcium chloride—and mechanical stabilization, which includes adding choice aggregates or utilizing geosynthetics to reinforce the soil.

MATERIAL TYPES



Type of Soil	Appearance / Feel	Water Movement	When Moist	When Dry
Granular soils, fine sands and silts	Coarse grains can be seen. Feels gritty when rubbed between fingers.	When water and soil are shaken in the palm of hand, they mix. When shaking is stopped, they separate.	Very little or no plasticity.	Little or no cohesive strength when dry. Soil sample will crumble easily.
Cohesive soils, mixes and clays	Grains cannot be seen by naked eye. Feels smooth and greasy when rubbed between fingers.	When water and soil are shaken in palm of hand, they will not mix.	Plastic and sticky. Can be rolled.	Has high strength when dry. Crumbles with difficulty. Slow saturation in water.

[THE IMPORTANCE OF MOISTURE CONTENT]

The importance of water to the process of soil compaction cannot be overstated. Every type of soil has physical characteristics that define how the soil reacts to moisture. For every soil material, there is a moisture content that maximizes the engineering properties of the material for a given compaction energy. Generally, the smaller the particle size, the greater the influence of water on compaction.



If a soil possesses too little moisture, the material will be difficult to work with because the particles will lack the lubrication necessary to rearrange them into a more dense state. And, the particles will not be cohesive enough to stay where they have settled.

Water is added to enhance cohesion and lubrication. Too much water can cause saturation. When the soil is saturated, voids fill with water, weakening the load-bearing capability of the structure. The particles will also be over-lubricated, allowing them to easily displace.

As a simple example of how water can affect the engineering properties of a soil, imagine trying to construct a sand castle on a beach. The water in the sand gives it enough cohesion to be molded into thick walls and tall towers. Now imagine trying to build the same sand castle in the desert. The arid desert soil lacks cohesion, so it would be difficult to manage more than a few low mounds of sand.



Water has other effects as well. Because it resists compression, it will displace soil materials, causing instability. Frozen water expands, displacing the soil around it. When the frozen water thaws, it occupies a smaller volume, creating space that will allow settling to occur.



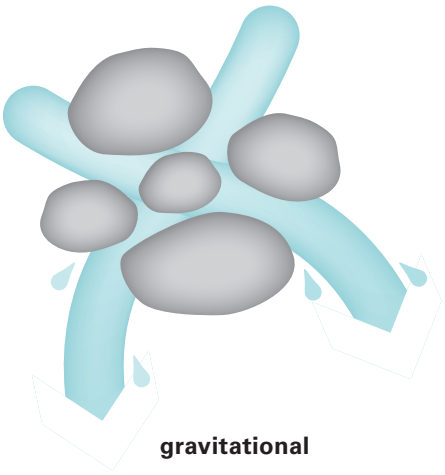
Water is present in all soils in their natural states. It appears in one of three ways.

1. Gravitational water is free to move downward due to the force of gravity. It can drain naturally from a soil.

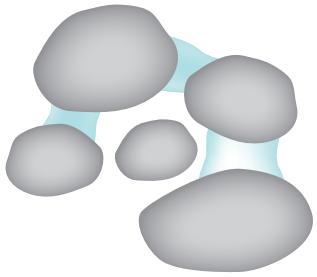
2. Capillary water is held in a soil by small pores or voids. It is considered free water and can only be removed by lowering the water table or by evaporation.

3. Hygroscopic water is present in a soil after gravitational and capillary water are removed. Individual soil grains hold this water in the form of a very thin film having physical and chemical affinity for the soil grains. It is also called "air-dry" moisture content. This water would have to be removed by baking the soil in an oven to determine the true dry weight of the soil.

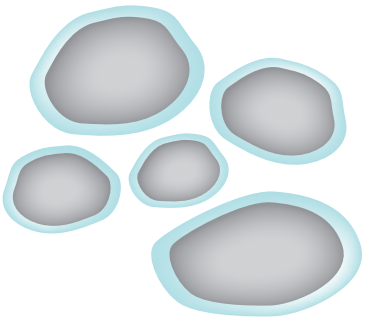
Too much moisture over-lubricates the soil and causes particle instability, while too little moisture reduces cohesion and prevents particles from reorienting easily into a denser state. For each soil type, there is an ideal moisture content at which maximum density can be reached with a given amount of compaction energy. The Proctor Test was developed as a way to help define this optimal moisture content for the selected compaction effort.



gravitational



capillary



hygroscopic



[THE PROCTOR TEST]

The value of compacting base and subbase soils has long been understood. But it was not until 1933 that Ralph R. Proctor of the Los Angeles Bureau of Water Works developed a standardized method for determining the optimum moisture content and the corresponding maximum dry density. The Proctor Test used a manually operated ram to compact three layers of the soil placed in a confined cylinder.

The Standard procedure determines the optimum moisture content of a material that will allow a given applied compaction force to obtain the maximum dry density of the material. This result is used to create a specification for compaction on the worksite. Because conditions in the field do not match the ideal conditions in a laboratory, the target compaction is scaled to a percentage of the dry density determined in the lab. This can range from 90 percent to over 100 percent.

Modified compaction tests have also been introduced in connection with structures requiring heavier bearing strength to support extremely heavy loads or to limit settlement. The Modified compaction test applies about four times more energy than the Standard compaction test and normally results in a lower "optimum" moisture content.



Proctor lab test kit

Surpassing 100 Percent Dry Density

How can the target density be over 100 percent? The maximum dry density established by the Proctor Test is not the maximum field density achievable on a particular soil. Proctor Dry Density of 100 percent represents the maximum density achieved in the lab with the particular sample using a specific amount of compaction force and ideal moisture content. Standard Proctor and Modified Proctor use different weights and will achieve different dry densities for the same sample. In the field, the "blows" are coming from a large soil compactor that applies a different amount of force than the hammers from the Proctor tests. It is not unusual to achieve field densities from 100 percent to 115 percent of Proctor maximum dry density. Geotechnical engineers may determine that due to load-bearing requirements and soil characteristics, compaction density over 100 percent of Proctor is warranted.

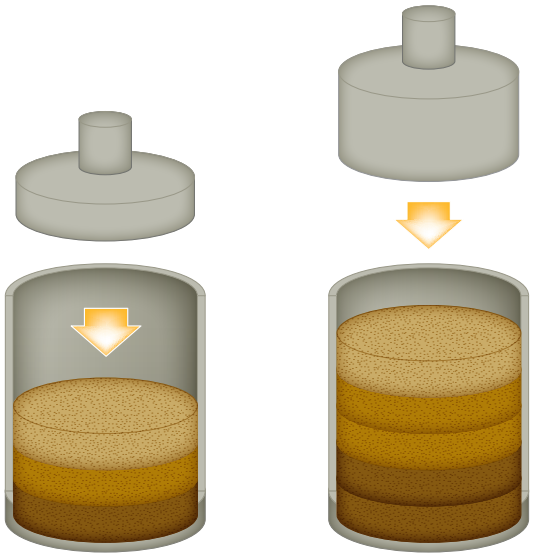
For a given soil sample, either the Standard or Modified Proctor is performed five times. The same procedure is used each time the test is run, but the moisture content is varied for each.

The series starts with the soil in a damp condition somewhat below the probable optimum moisture content. After the first sample is compacted into a cylindrical container, the wet weight is measured and a portion of the sample is placed in a drying oven. When the sample is completely dry, it is weighed again. The difference between the wet and dry weights yields the moisture content that is expressed as a percent of the dry weight.

A second sample with increased moisture content is compacted and the weighing and drying process is repeated. Additional samples with increasing moisture content are processed until the wet unit weight decreases or the soil becomes too wet to work.

Dry density and moisture content values for each sample are then plotted and a smooth curve is formed. The highest point on the curve represents the maximum dry density and the optimum moisture content for that soil sample.

PROCTOR TESTS



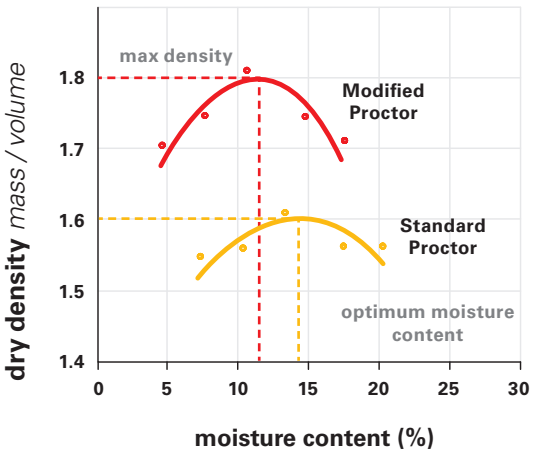
standard

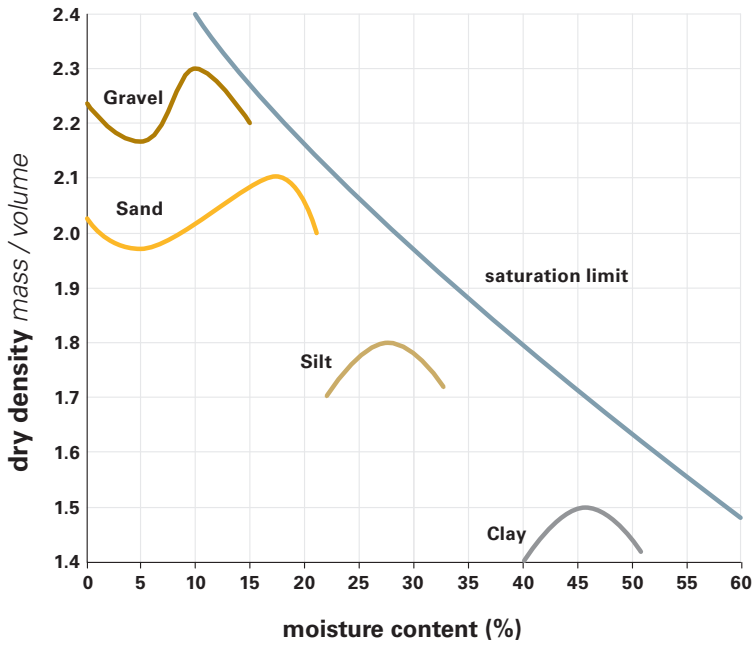
Each layer receives 25 blows from a 2.5 kg (5.5 lb) hammer at a distance of 305 mm (12 in)

modified

Each layer receives 25 blows from a 4.5 kg (10 lb) hammer at a distance of 457 mm (18 in)

PROCTOR CURVES

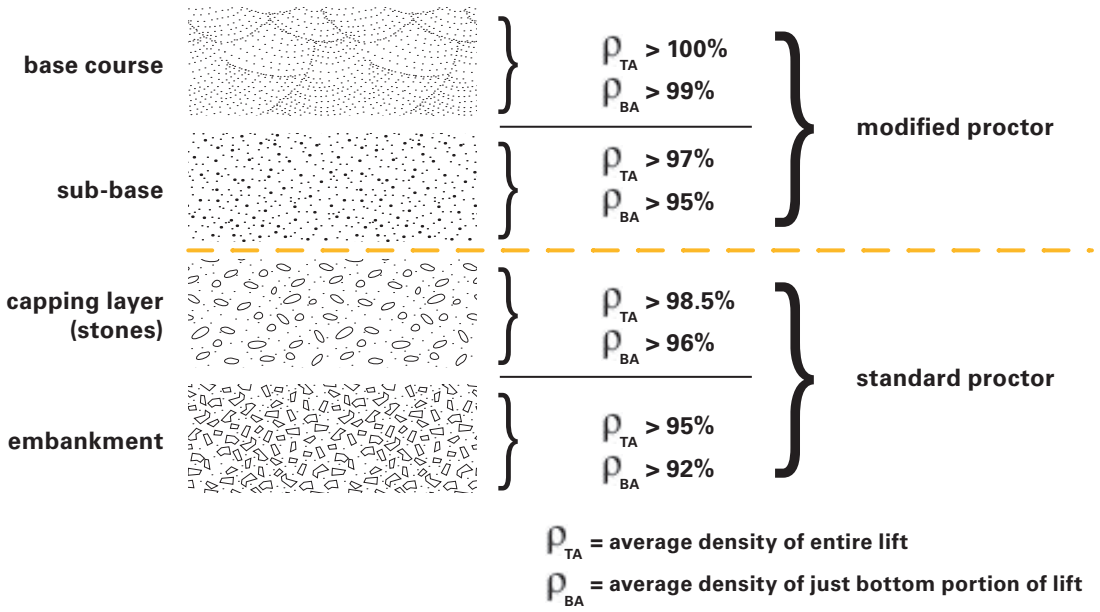




Laboratory tests determine the moisture content at which maximum density can be attained for that particular soil material. Field target densities are specified as a certain percent of the maximum laboratory dry density. Generally, required field densities will be 95 percent of Standard Proctor for embankments and up to 100 percent of Modified Proctor for roadway structures. Likewise, the moisture content must be within a range of the laboratory-determined, optimum moisture content.

DENSITY TARGETS

This example shows how the nearer the material is to the surface, the higher the density.



This illustration of the gradient of compaction compares the average density of the entire lift (ρ_{TA}) with the average density of just the bottom portion of the lift (ρ_{BA}).

